

Description

This application note aims to explain the fundamentals and motivation of using a system with Power over Data Line capabilities. Some of the advantages of using a system with these capabilities will be shown. While additional circuits will be required to enable a system to use this, different circuits with different properties including complexity and cost will be shown to help in deciding which solution will be required. Due to the inherent complexity of this feature, simulations are recommended and the results of simulations will be compared to their corresponding measurements.



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1. Preface

1.1. General Terms

Table 1-1. General Terms

Term	Description
10BASE-T1L	10 Mbit/s Ethernet over long-reach single pair of conductors, IEEE Std 802.3 Clause 146
10BASE-T1S	10 Mbit/s Ethernet over short-reach single pair of conductors, IEEE Std 802.3 Clause 147
BIN	Bus Interface Network
CMC	Common Mode Choke
DMC	Differential Mode Choke
PD	Powered Device
PoDL	Power over Data Line
PoE	Power over Ethernet, IEEE Std 802.3 Clauses 33 and 145
PSE	Power Sourcing Equipment

1.2. Reference Documents

1. AN1829 - Understanding System Inherent Noise: Enabling Longer Reach and More Nodes on 10BASE-T1S Systems

Available on the Microchip website, listed under the LAN8650, LAN8651, LAN8670, LAN8671 product pages.

www.microchip.com/DS60001829

2. An Introduction to Power over Data Line

Reducing the costs of components and assembly is an important goal in most hardware designs, regardless of the end market. Power over Ethernet (PoE) enables system designers to achieve this goal by using a single cable to both transfer data and power the device. PoE reduces assembly and material costs, since only one cable must be laid out and connected. It also reduces weight, which is especially important in automotive and other transportation applications.

PoE was initially developed for Ethernet systems using CAT5 or CAT6 Ethernet cables, which contain 4 pairs of wires, and PoE implementations use multiple pairs in these cables. The goal of reducing cost and weight has driven the development of Ethernet standards that require only a single pair of conductors, such as 100BASE-T1, 10BASE-T1L and 10BASE-T1S. Older PoE standards were not designed for only a single pair of wires; Power over Data Line (PoDL) is the solution when using a single pair.

Systems using PoDL are split into two main components: the Power Supply Equipment (PSE) and the Powered Devices (PD). As the name suggests, the PSE is responsible for providing power to the devices connected on the transmission line. The PD is the device being powered by the PSE. PSEs can be described in terms of their voltage, current and quality of provided power. In all cases, PSEs must be able to endure the in-rush current that loads the transmission line and the PD. PDs must have a circuit to separate the power supply from the data.

PoE standards on a point-to-point network often include a negotiation protocol to ensure necessary voltage and current are provided. On engineered systems, where the devices on the bus are already known during the design phase, it is possible for the PSE to omit the communication phase. On such systems, the PSE may power the device to the predetermined power class. This is especially useful in systems used in the automotive industry because it saves time during startup.

The IEEE 802.3 working group has defined the standards for PoDL for several point-to-point single pair Ethernet variants, such as in the IEEE 802.3bu and IEEE 802.3cg standards. The IEEE 802.3cg standard defines PoDL for both 10BASE-T1L and 10BASE-T1S *point-to-point* segments. One of the key advantages of 10BASE-T1S is the ability to have more than two data nodes on one mixing segment. This makes the mixing segment a switch and enables many low bandwidth devices to communicate via standard Ethernet. At the time of creation of this document, there are no PoDL standards for 10BASE-T1S multidrop mixing segments. For these networks, inspiration from other applications can be found and used instead. This can include applications based on PoE, 10BASE-T1L systems, systems using point-to-point links, RS-485 based systems, or even USB specifications. Because there are no tight constraints to uphold, every solution to PoDL for multidrop systems is potentially different and unique.

The main challenge when developing a system using PoDL on a 10BASE-T1S multidrop mixing segment is that there is no standard defined for this case. The standards listed above define PSEs of differing power classes as well as negotiation procedures to assure safe operation between PSE and PD before full power is applied. The solutions proposed in this application note are therefore not adherent to a specific standard. Nonetheless, the solutions developed using this starting point will still be usable in engineered systems in the future.

This application note will show examples of PoDL circuits that have been tested with the LAN8651, LAN8670 and LAN8671, either of which can be used as a starting point for custom designs. It is assumed that the mixing segments shown and developed here are part of an engineered system, with the number and type of PD known beforehand and compatible with the PSE. The PSE is able to supply enough power for all active PD as required.

3. Evaluation Setup

While the IEEE PoDL and PoE specifications define the behavior characteristics required for a point-to-point link segment, multidrop networks are not covered in these specifications. For these networks, inspiration from other applications can be found and used instead. This can include applications based on PoE, 10BASE-T1L systems, systems using point-to-point segments, RS-485 based systems, or even USB specifications. Because there are no tight constraints to uphold, every solution to PoDL for multidrop systems is potentially different and unique. In this case, it is necessary to define the constraints placed on the characteristics of the PSE, the PD, and even the characteristics of the multidrop system itself. The proposed system used in the evaluation tests will be described in the following sections.

Evaluation Power Supply Equipment

Figure 3-1. Outline of evaluation Power Supply Equipment

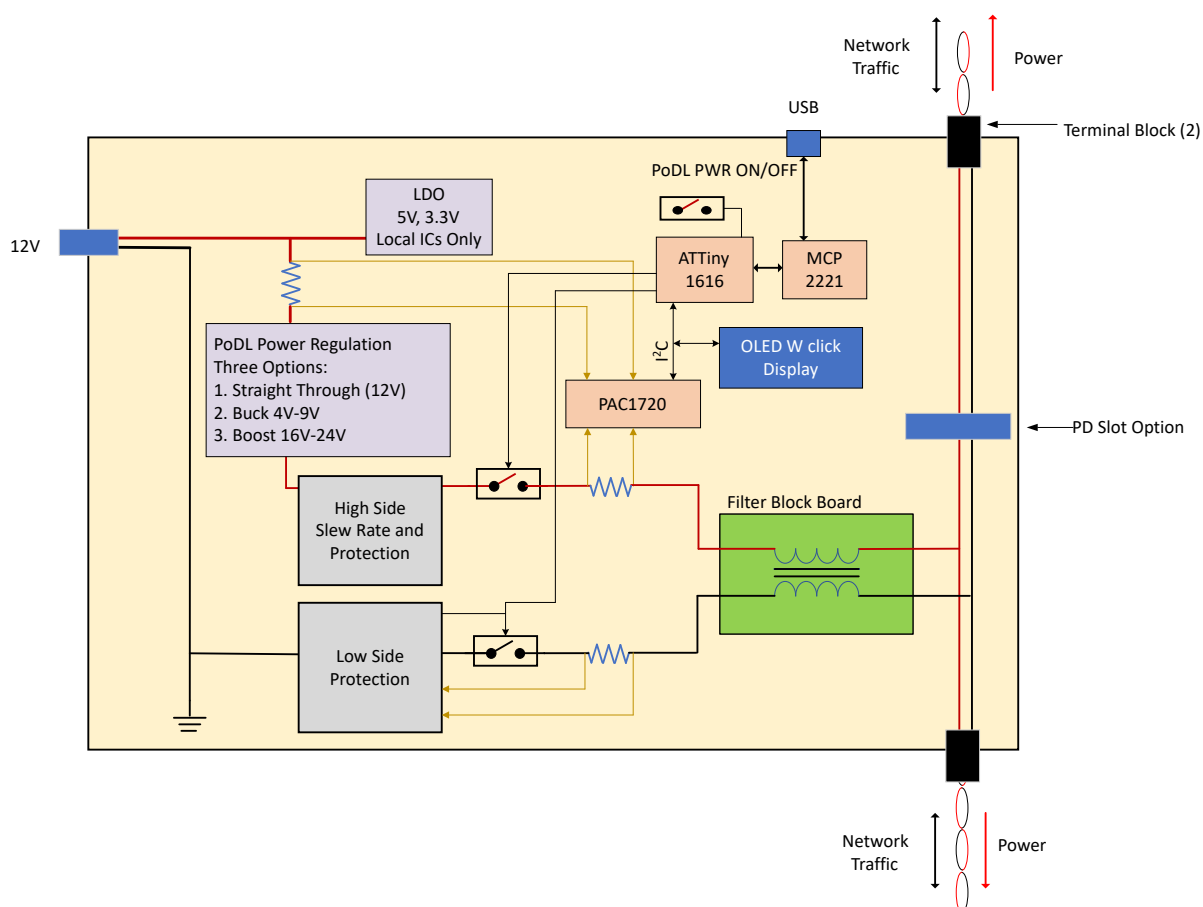


Figure 3-1 shows a high level block diagram of the PSE used for the evaluations. For the power source, a laboratory power supply with a stabilized 12V output was used. When different input or output voltages are required, it is possible to use a switching regulator such as a buck or boost converter. One thing to note is that if a switching regulator is used, it will potentially cause voltage ripples and produce additional noise on the output voltage. These ripples will contribute to overall system noise and should be filtered before being injected onto the network.

The green function block in Figure 3-1 above is implemented as an add-on board. This allows for easy testing and swapping of different power supply filter options. Three options will be evaluated: one filter uses a Differential Mode Choke (DMC), one uses two coupled inductors, and one uses two

separate uncoupled inductors. The effectiveness of the filters will be analyzed by comparing these different filters to each other.

Another important detail to note is that the PSE contains no physical layer connection for data transmissions on the network. The PSE will be used only to power the equipment on the PDs. The effects of including a PHY locally on the PSE are not taken into account here.

Transmission Line Cable

The cable used to transmit the data and power is a simple single twisted pair cable. This particular type of cable is often used in Microchip demonstrations and trainings. The cable used in testing is Interstate Wire WIA-2207-0/3, which is a 22 AWG wire, with a temperature rating of up to 80°C and voltage rating of up to 150V. The wall thickness is 0.010 inches and with two twists per inch. Characterization tests of the cable showed that the DC resistance of the cable is 0.46 Ω /m and the propagation delay is 6 ns/m.

Network Topologies

The topology of the network is varied between subsequent tests and simulations. This helps test the system in different configurations. It shows whether the underlying system works in different situations or if it only works in specific situations. In the tests and simulations, four different topologies will be used. To prevent effects of node clusters affecting the network, it is recommended that nodes be placed at least 15 cm apart. Additionally, the PSE is located on the physical end of the transmission line for the topologies shown in [Figure 3-2](#) below.

Figure 3-2. Topologies used for testing and simulation

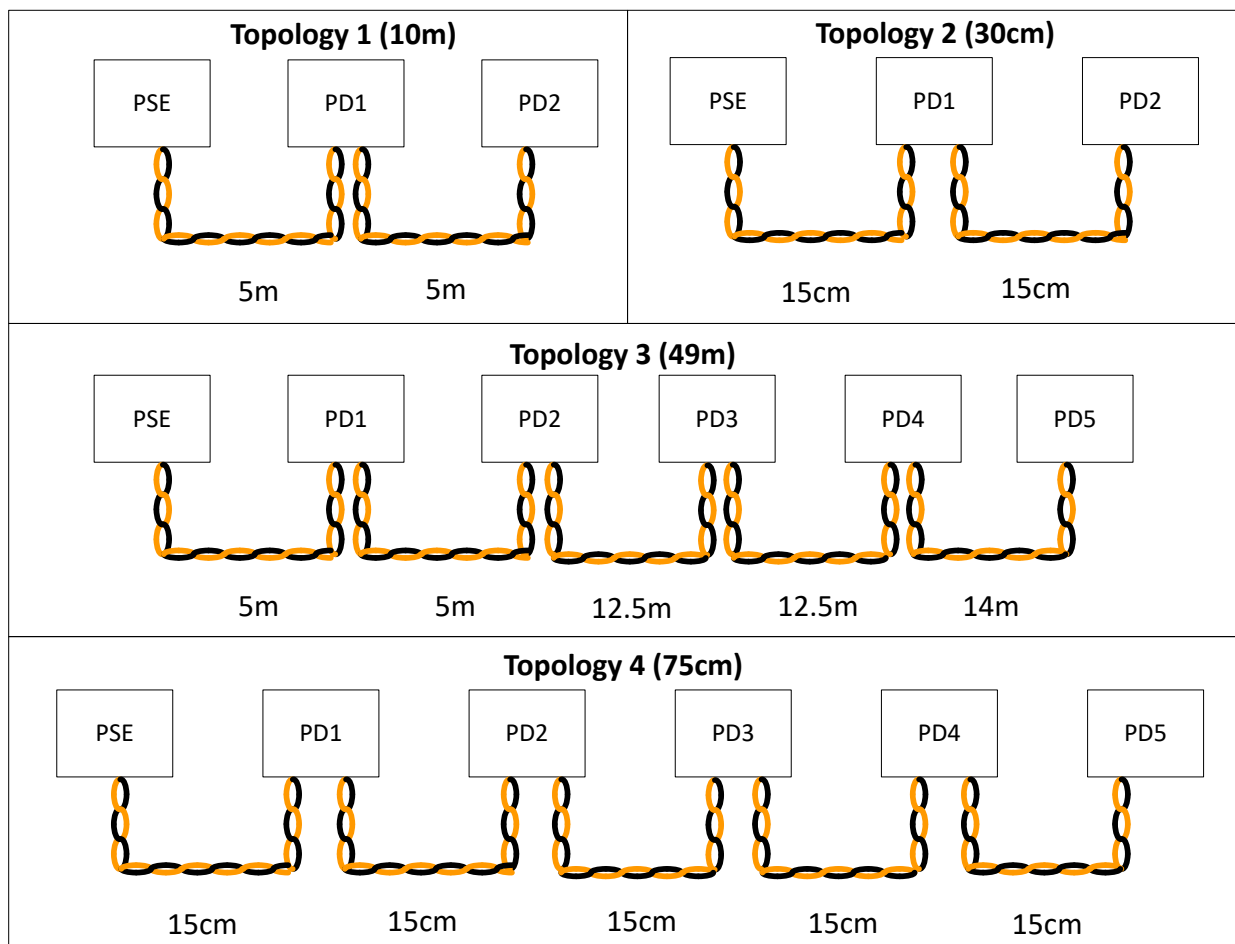
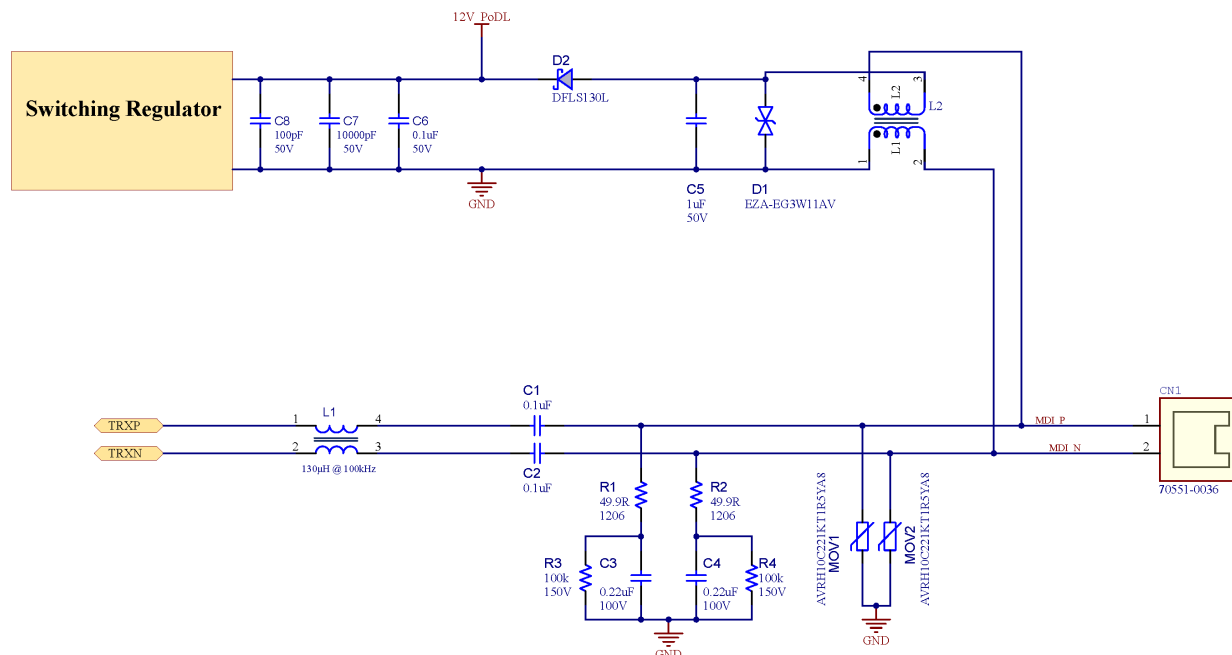
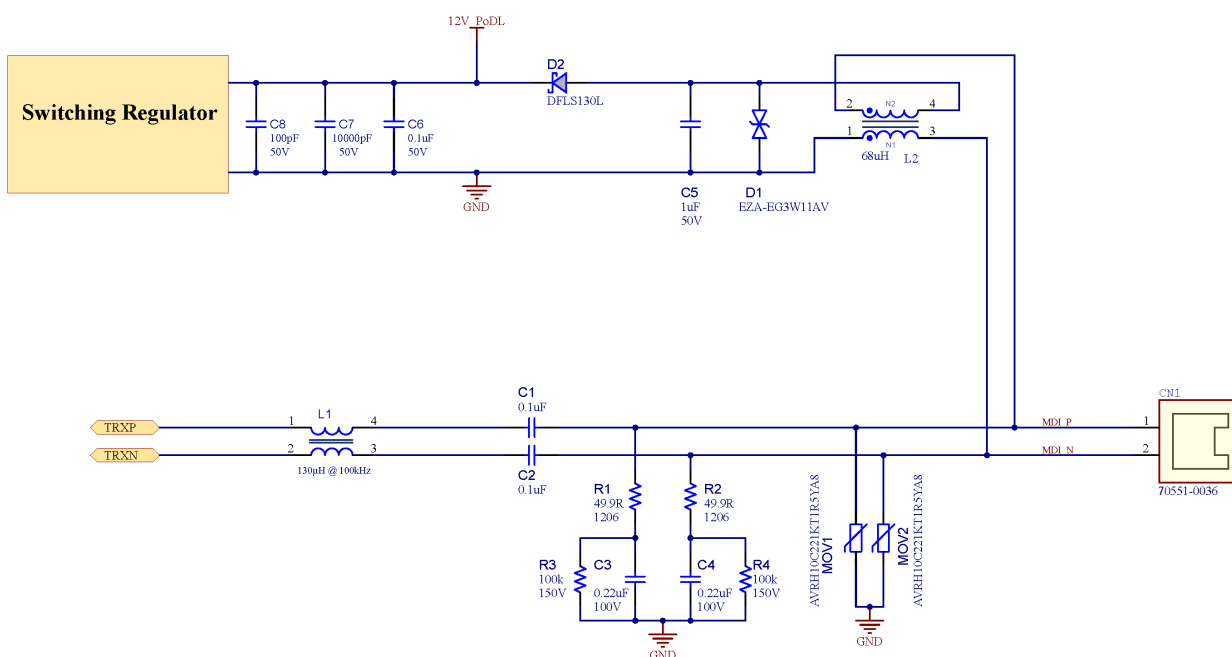


Figure 3-4. Schematic of linked inductors filter solution



Finally, it is possible to use strongly coupled inductors, such as a Differential Mode Choke (DMC). Common mode Chokes (CMCs) can often be configured to work as DMCs. This is the most expensive solution, but it also yields the best signal quality and therefore the best noise margin for data on the transmission network.

Figure 3-5. Schematic of DMC filter solution



The coupling and filtering between the PD and the network greatly affects the noise performance of each device as well. Since there are multiple devices using a similar coupling and filter, bad performance of the filter may compound the performance and reduce overall noise margins. From

the point of view of the PSE, each coupling on the PD is parallel to each other. This means that for an increasing number of nodes, each additional coupling adds a further inductor parallel to the others, reducing the overall inductance of the transmission line as seen from the PSE.

Regardless of the filter type used for the power supply input on both the PSE and PD devices, a key characteristic of each device is the behavior of its power supply input. Often times, devices will also require a regulator to transform the variable voltage from the transmission line into a stable 1.8V or 3.3V supply. However, a regulator may also cause ripples and reflections to couple back into the network. To measure this, a good testing point is shown in [Figure 3-3](#) and [Figure 3-5](#) above. The testing point is between the locations marked as *12V PoDL* and *GND*. The top left side of the schematic above in [Figure 3-5](#) will be used as an input to a power converter to reduce the voltage to the required level for the 10BASE-T1S transceiver. The filter schematic of the power converter is part dependent and will not be shown. As a reference, when powering the device using a 12V battery and using a DMC based power filter of 47 μ H, a differential noise of 91 mV and common mode noise of 43 mV were measured between the locations marked as *12V PoDL* and *GND*. The noise was measured again on the output of the switching regulator powering the 3.3V supply voltage. This was measured as 34 mV differential noise and 29 mV common mode noise.

The number of connected devices becomes important when deciding whether to use a capacitive or inductive coupling on the Bus Interface Network (BIN) of the PD to filter out the power supply and connect data to the Ethernet ports. Capacitive coupling uses a CMC and capacitors to filter out the lower frequency power transmissions. Inductive coupling uses a transformer between the PSE network and the device to filter out the power supply part of the signal. In both cases, only higher frequency signals are allowed to reach the device data ports, while DC and inherent low frequency standing waves are used to power the device.

In the evaluation tests, the coupling and BIN used are based on capacitive coupling. The BIN provides a capacitor in series to the TRXP and TRXN pins and a CMC across both lines. The capacitance is very small in the BIN, with the dominating factor being the inductance of the CMC.

4. Evaluation Simulations

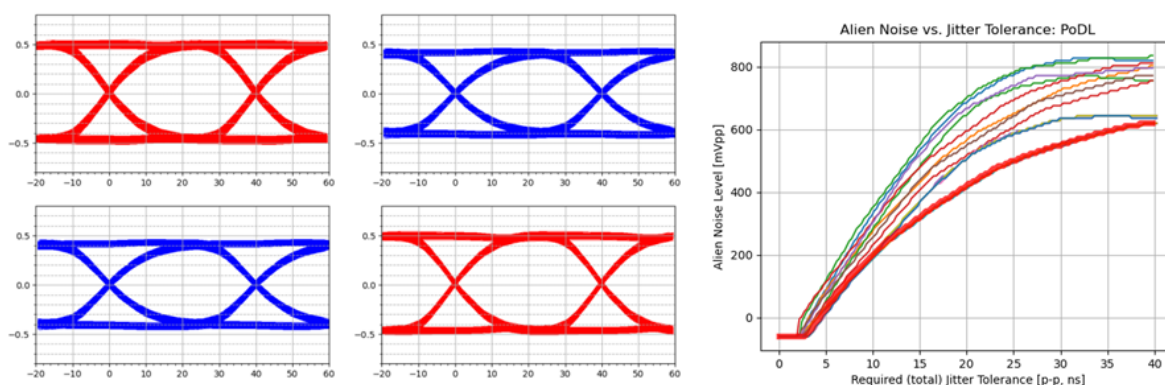
To evaluate the effects of different configurations, placement of components and component values, various schematic variants with the three coupling filters presented earlier were tested using simulations. The setup for the different testing scenarios were derived from previously available system level templates used for data evaluation. The Keysight ADS software was used to perform these simulations. The power supply coupling filters were added to each configuration and the properties and characteristics of the inductor arrays, weakly coupled inductors, and DMC were taken from the respective manufacturers.

For each of these filters a schematic was created and simulated. Not only was the type of filter tested, but the values of the inductance was also varied to see which values lead to the best performance. As a reference, the system was also simulated without using PoDL to see what effects the power transmission has on the signal quality. The results are visualized using the eye diagram of each node of the simulation and the corresponding alien noise and jitter tolerance plot.

Since there were many simulations done during the initial phase, only the most relevant results will be discussed here. An example using only two nodes on Topology 1 and an example using five nodes on Topology 3 will be shown. The example on Topology 3 will be used as a worst case scenario, the resulting eye diagram and jitter plots will be discussed.

Figure 4-1 below shows the results from a simulation consisting of two nodes and a power supply. The coupling between the devices and the network is realized using single, uncoupled inductors with a nominal value of $47\ \mu\text{H}$ on both the positive and negative wires of the transmission line. The resulting eye diagram and jitter plot show that a large noise margin is left for the system to utilize.

Figure 4-1. Simulation using two $47\ \mu\text{H}$ single inductors on Topology 1

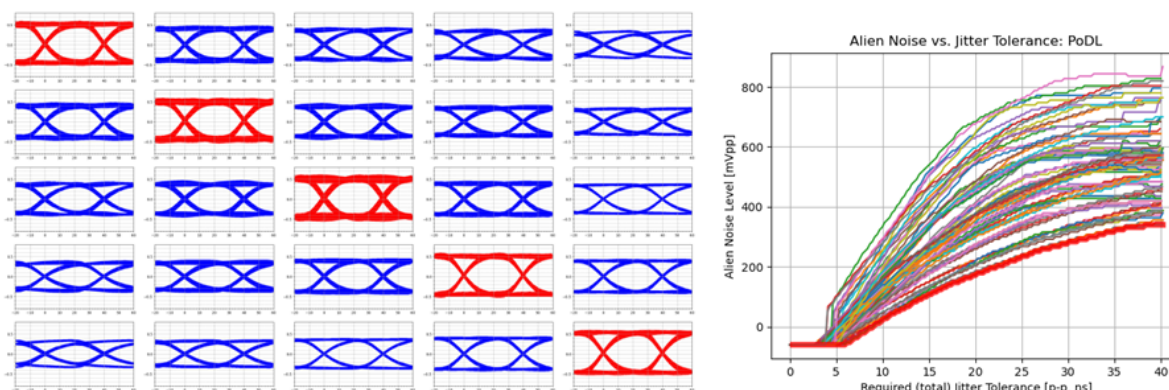


NOTICE

The noise and jitter plot contains the same information as the eye diagram. This plot is the superposition of each quadrant in the eye diagram normalized to show a single edge. This can also be achieved by splitting both the X- and Y-Axis into two separate parts and rotating the result to show a rising edge. The alien noise and jitter tolerance plot can be used to quickly see the worst case scenario for all participating nodes in the measurements. This is the lowest line in the plot, highlighted with red. An offset of 60 mV is subtracted from each measurement to account for the ± 30 mV signal detection threshold of the device. The y-intercept can be interpreted as the amount of jitter that is created by the system itself. This is the jitter in the system if there is no alien noise present. Since the maximum allowed jitter is 15 ns, the value of the red line at a jitter value of 15 ns indicates how much noise tolerance the system has in total. This is the amount of noise that the system may be subjected to before communications fail. If this value is below zero, it means that the system will contain at least one path failing to communicate correctly, even without external noise applied.

Figure 4-2 below shows a simulation of five nodes and a separate power supply. Here, each node uses a coupling to the transmission line consisting of two 68 μ H inductors, one for the positive and one for the negative wire. The resulting eye diagram and jitter plot are shown.

Figure 4-2. Simulation using 68 μ H single Inductors on Topology 3



When comparing the two results, it is obvious that the five node configuration is inherently much more sensitive to external noise than the two node configuration. This is because the topology of the system itself causes noise on the network, reducing the overall noise threshold at which communication between nodes will be disrupted. However, the noise performance is still within acceptable bounds because there is a large margin before additional noise becomes an issue, meaning that every node is able to both receive power over the transmission line and still communicate successfully with each other node in the system.

Figure 4-3 below shows the topology used to test an additional configuration with eight nodes. In this configuration the PSE was moved from the end of the cable to a location between the fourth and fifth nodes. The filter between the power supply on each device and the network was realized using a 47 μ H coupled inductor on each device.

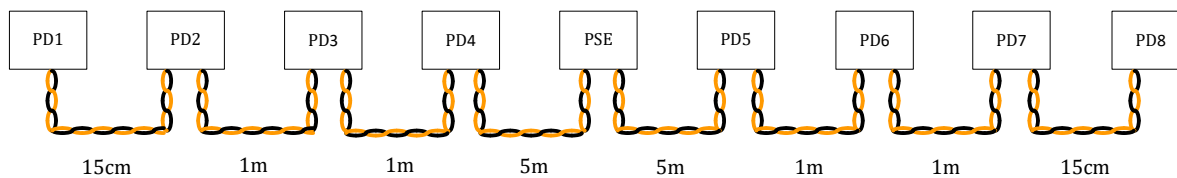
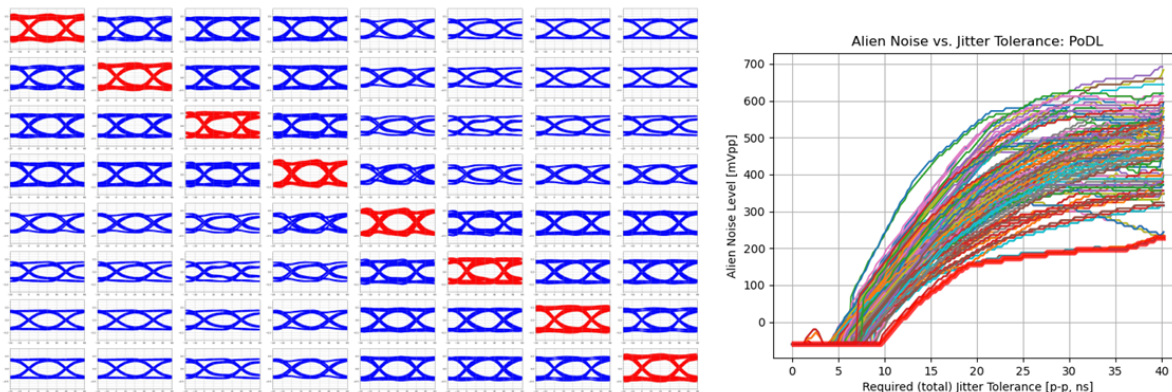
Figure 4-3. 8 Node Topology

Figure 4-4 below shows the results of the simulation of the eight node topology. The overall noise tolerance of this system is the lowest so far, but for environments with lower alien noise it is still possible to implement this system successfully.

Figure 4-4. 8 Node Topology Simulation

5. PoDL Evaluation Measurements

To verify the results of the simulations and to check the behavior of the systems in a real environment, the evaluations were repeated using real hardware in a laboratory environment. The same topologies and couplings were used, and the same values for the components were used. The results of the simulations will be compared to the results of these measurements.

Figure 5-1 below shows the results from the measurement of the same configuration that was simulated previously. The resulting eye diagram and jitter plot of the measurements are nearly identical to the simulation results.

Figure 5-1. Measurements using two 47 μH single inductors on Topology 1

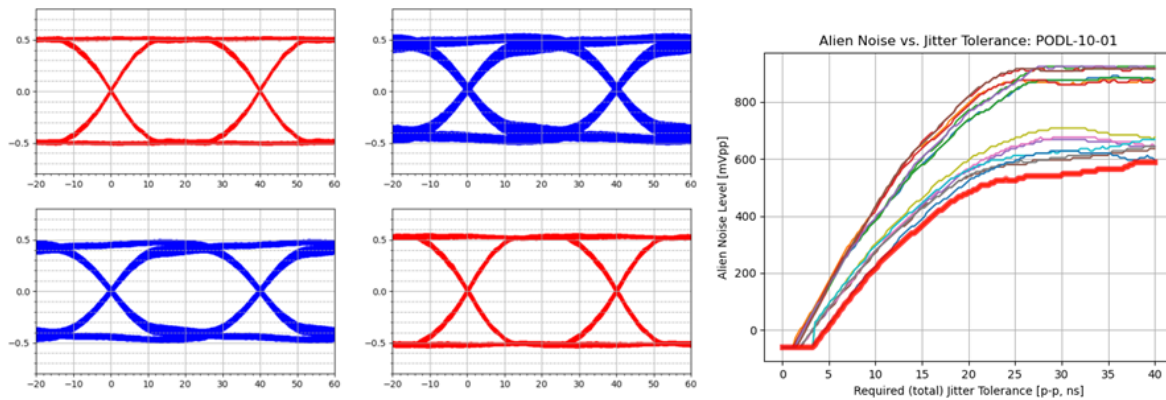
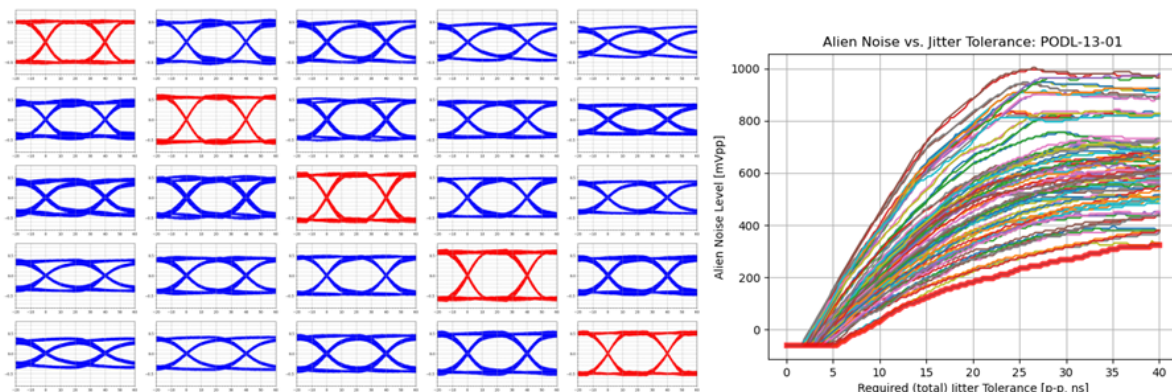


Figure 5-2 below shows the results of the measurement of the second system that was simulated. Again, the resulting eye diagram and jitter plot are not exactly equal, but still very close to the results of the simulation.

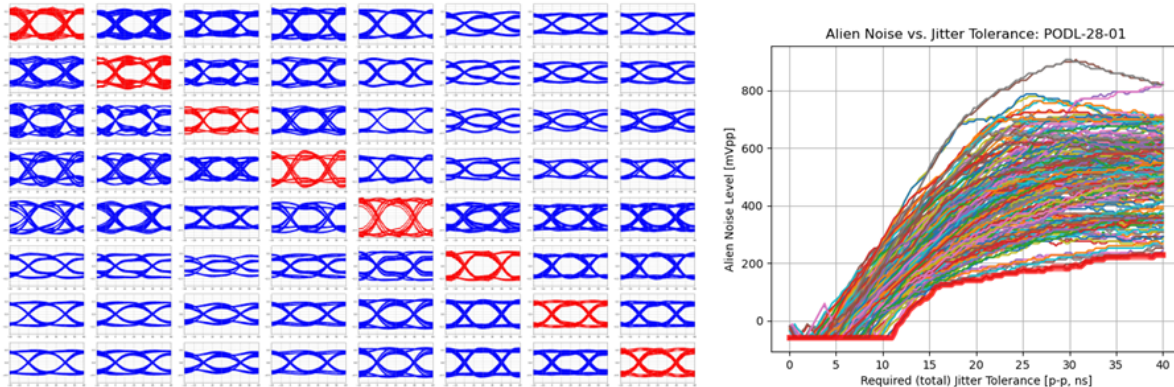
Figure 5-2. Measurements using two 68 μH separate inductors on Topology 3



In both the simulation and measurement, the second system shows much worse noise and jitter performance than the first system. However, despite this, both systems are fully functional. The second example is used as a worst case analysis of the results. An important thing to note is that the amount of external or alien noise will greatly affect the system. In a high noise environment, a high alien noise tolerance is required for the system to work properly. On the other hand, if the system is employed in an environment with little external noise, it will still remain functional even with a very low noise tolerance. Therefore, to make a valid prediction about the functionality of the system the level of alien noise must also be known.

Finally, the setup for an eight node system was tested. The topology is the same that was simulated previously. The results of the eight node measurement are shown below in [Figure 5-3](#). The Alien Noise and Jitter Tolerance plot shows that the system has some noise tolerance left over. If the environment the system is placed in does not have too much noise, it will function correctly.

Figure 5-3. Measurements using 47 μH coupled Inductors on 8 Node Topology



[Table 5-1](#) below shows some of the inductors used in testing. One of the biggest issues when using single inductor pairs is finding equally matched inductors for each device. Using matched inductors is recommended to keep the symmetry of the differential line. If different value inductors are used, the eye diagram for the particular node will not be symmetric, reducing noise margin. Furthermore, the inductors used are only rated for a relatively low current. This may be acceptable for each PD, since it requires little power individually, but this will become an issue for the PSE when it will need to power each device simultaneously. Measurements and tests also showed that the tolerance of the components can play a major role on the workability of the solution. A minimum inductance on each node is required. If the actual inductance falls below this because the components tolerance is too high, then the system might be more susceptible to external noise.

Table 5-1. Component Used during Testing

Architecture	Nominal Inductance (μH)	Saturation Current (mA)	Rated Current (mA)	DC Resistance ($\text{m}\Omega$)	Tolerance (%)
DMC	47	300	250	806	± 20
Single Inductors	15	1100	850	252	± 20
Single Inductors	47	700	-	660	± 20
Single Inductors	68	760	-	420	± 20
Coupled Inductors	47	1375	900	300	± 20
Coupled Inductors	68	2220	1220	230	± 20
Coupled Inductors	47	-	1440	174	± 20

Power Consumption

While taking the measurements during the tests described above, the connected nodes were powering the local microcontrollers. While not measuring the power consumption of each individual node, the total power consumption of the system was measured instead. The results are shown below in [Table 5-2](#).

Table 5-2. Measured Power Consumption of System Configurations

Number of nodes	Network Supply Voltage	Total current
2	12V	160 mA
5	12V	480 mA
8	12V	740 mA

Additional measurements were done for the eight node system. For this system, a load variation test was conducted. During this test, one node draws additional power to simulate cyclic power consumption of an application or transmission. To test this, a dynamic load was applied on the low voltage side of the power regulator. This dynamic load was implemented by adding a $13\ \Omega$ resistance parallel to the rest of the node. This extra load resembled a rectangular signal with a variable frequency. The frequency of the signal was changed to the values shown below in [Table 5-3](#). During these tests, the system was monitored to see if this has an impact on the noise created in the system and if communication between nodes is still possible. During any of these tests, no communication was lost and the nodes were able to transfer data as normally. No major impact on noise behavior was detected. The results of these tests are shown below.

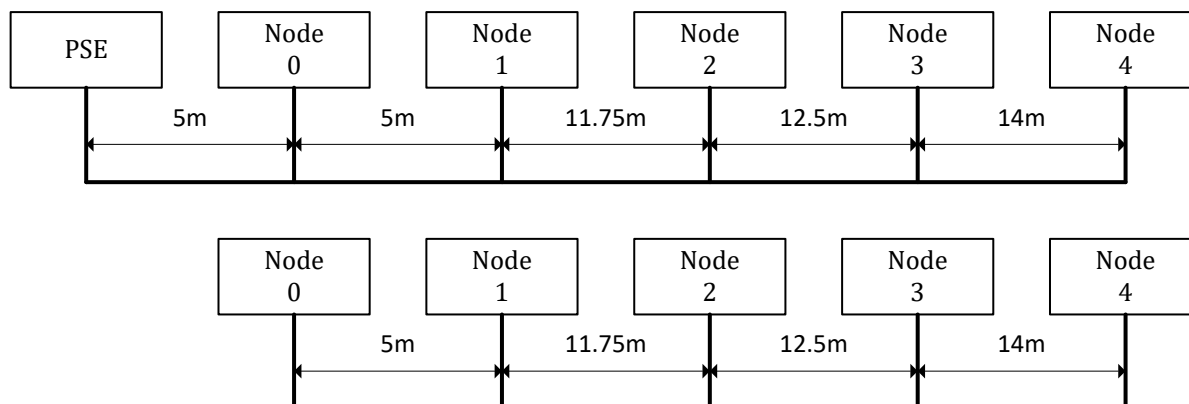
Table 5-3. Load Variations for Eight Node System

Load Variation Frequency	Network Supply Voltage	Total Maximum Current
77 Hz	12V	955 mA
778 Hz	12V	962 mA
180 kHz	12V	1004 mA

6. Example Applications

To show the advantages and disadvantages of using Power over Data Line on a mixing segment, a practical example with several PD nodes and a single PSE was created and tested. This mixing segment was then tested and measured once while PoDL was used, and once where the PoDL power supply circuit was missing on the PCBs. In this case, the cable to the PSE on the network was also removed. The topology used is shown in [Figure 6-1](#).

Figure 6-1. Practical Example Topology



The PSE consisted of a simple injector circuit which measured the injected voltage, current and total power. The power output was filtered by a pair of coupled inductors with a nominal inductance of 47 μ H and a maximum current of 1.44A. The voltage injection of the PSE was measured at 11.50V with a voltage noise of 1.6 mV Root-Mean-Square(RMS) and 22.6 mV Peak-to-Peak(PkPk) when measured across a 47 Ω resistor, providing 2.8W of power.

To measure the amount of noise on the mixing segment due to switching regulators and LDOs, a quick measurement was done while all devices were put into sleep mode. In this case, the PSE supplied a voltage of 11.55V at 2.0W with a RMS noise of 3.2 mV and a PkPk noise of 81.1 mV.

A similar measurement was done while all devices were active and powered. The PSE provided a voltage of 11.44V at 3.5W, with a RMS noise of 3.5 mV and PkPk noise of 78.3 mV.

The PDs used the same hardware and only differed in their Node ID. The schematic of the BIN circuit is shown in [Figure 6-2](#), while the layout of the BIN is shown in [Figure 6-3](#). The jumpers J14, J15 and J16 are used to connect a separate add-on board containing the PoDL circuitry. Jumper J14 connects the T1S PoDL network to the input circuitry of the board, jumper J15 connects the 5V power supply level to the add-on board and jumper J16 connects the ground GND to the add-on board. The add-on board uses the PoDL supply to generate a 5V output, supplying the main board with power. When used, the add-on board provides power to the main board, when omitted the main board must be powered by an external power supply.

Figure 6-2. PD BIN Schematic

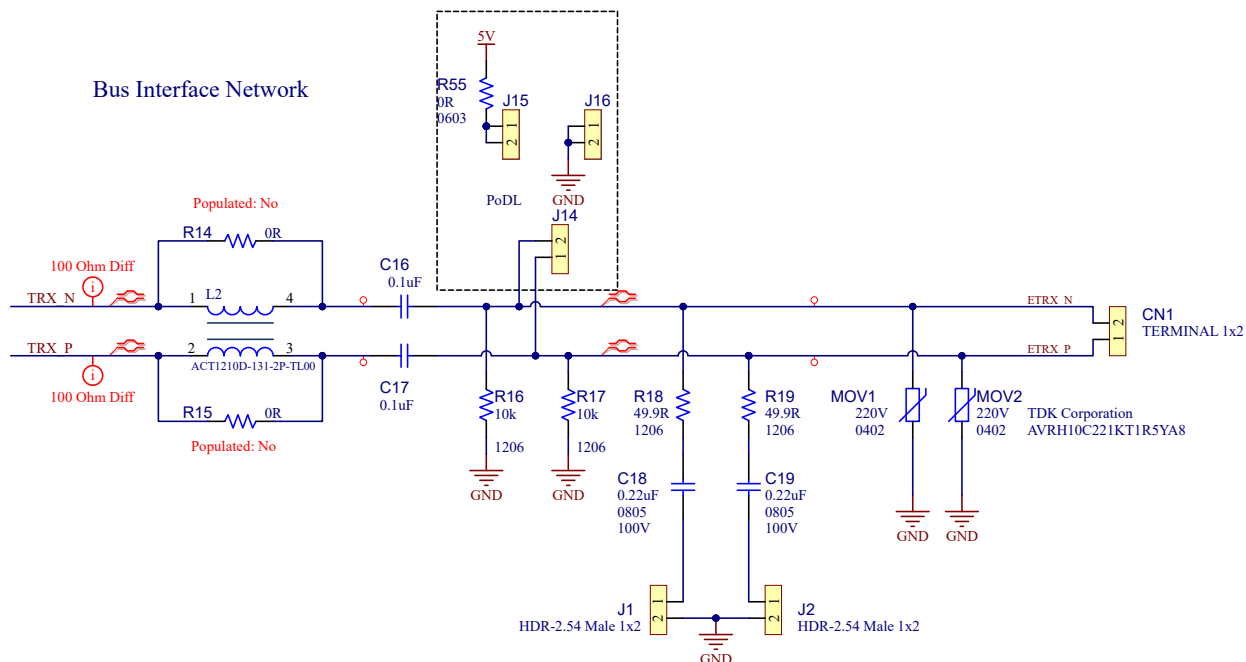
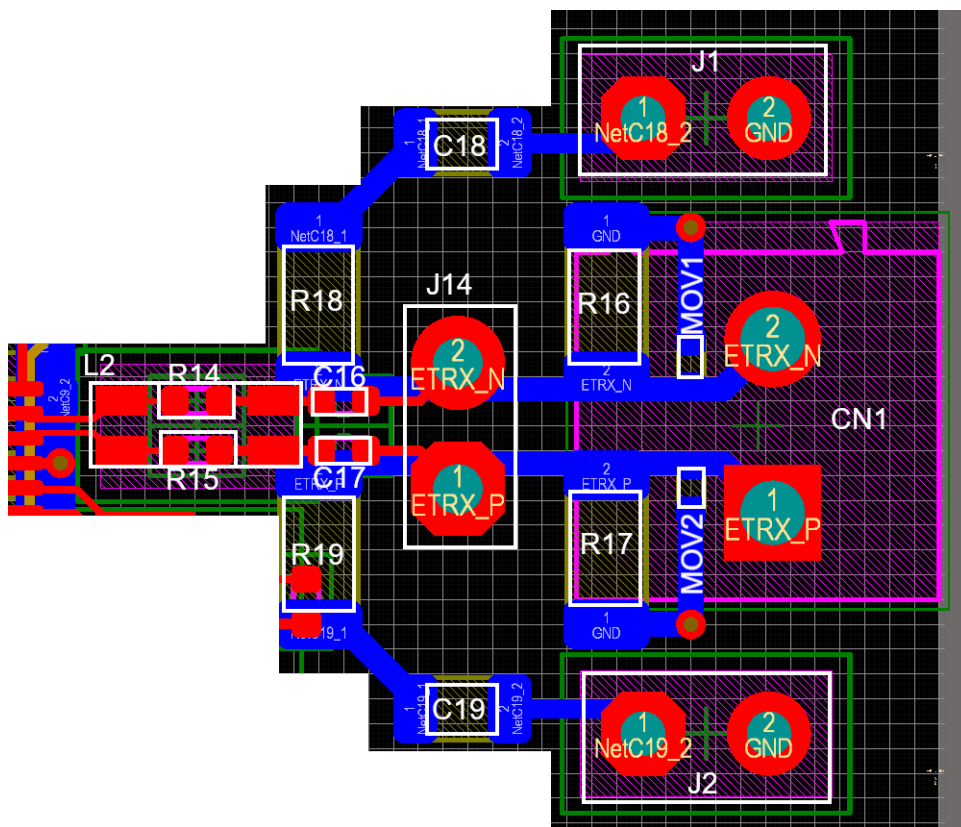


Figure 6-3. PD BIN Layout



An add-on board was used to provide the PoDL circuit. The schematic of the add-on board is shown in Figure 6-4, while the corresponding layout is shown in Figure 6-5. Jumper J1 connects to a T1S network with PoDL. Jumper J2 is a 5V power supply output, powered by the PoDL network, while

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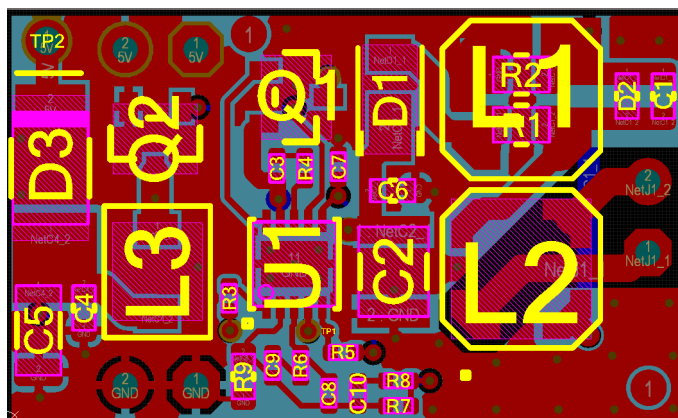


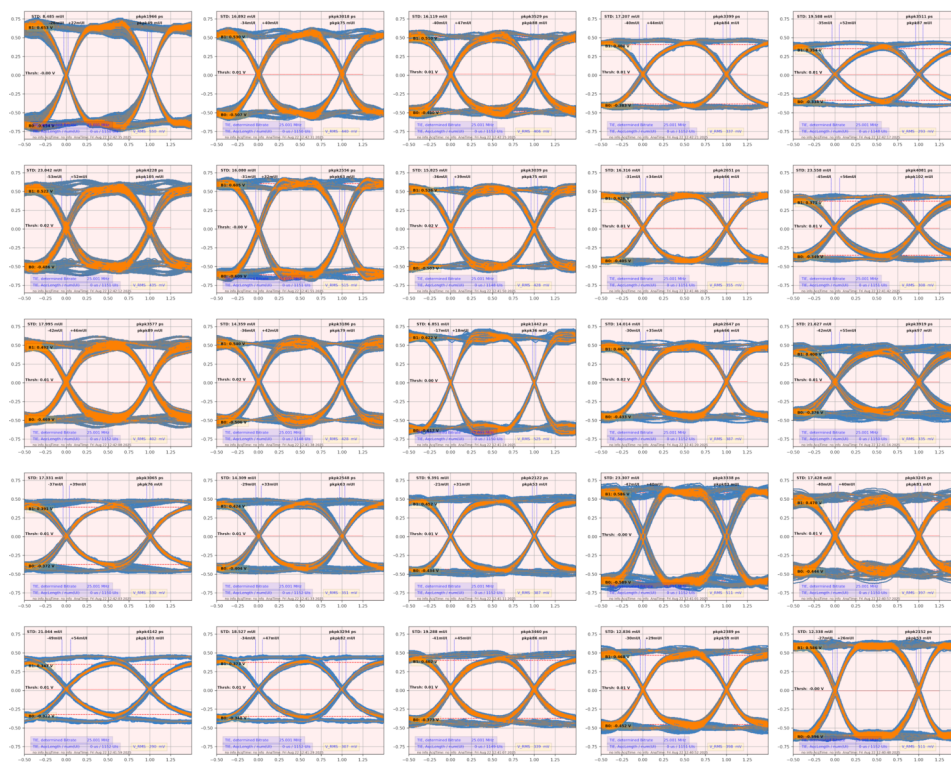
Figure 6-5. Add-On PoDL Layout

6.1. Example Application with PoDL

The example application uses five nodes with the same hardware on each node. The PoDL power circuit is provided by the add-on board shown earlier.

The following eye diagram measurements were done on the TRXN and TRXP pins of a LAN8651. These measurements were done for each combination of transmitter and receiver (i.e. Node 0 transmits and Node 4 receives, Node 0 receives and Node 4 transmits). The resulting eye diagram matrix is shown below in [Figure 6-6](#).

Figure 6-6. Eye Diagram Matrix for PoDL Setup



It is clearly visible that the worst case scenario for this mixing segment is when Node 0 transmits and Node 4 receives, or vice-versa. In either case, a jitter of about 2 ns exists on timing of signals with a maximum eye-opening of about 402 mV and a signal noise of about 5.0 mV.

6.2. Example Application without PoDL

The example application uses five nodes with the same hardware on each node. The hardware used is the same as with the PoDL setup, with only the PoDL power supply boards missing. Power is instead supplied directly to the boards using USB cables.

The following eye diagram measurements were done on the TRXN and TRXP pins of a LAN8651. These measurements were done for each combination of transmitter and receiver (i.e. Node 0 transmits and Node 4 receives, Node 0 receives and Node 4 transmits). The resulting eye diagram matrix is shown below in [Figure 6-7](#).

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First, there was no significant change in the shape of any of the eye diagrams. The general shape and outline as well as any effects caused by reflections did not seem to change between the tests. This means that when adding PoDL to a system, it should be expected that the eye diagram does not change significantly.

The increased noise may be caused by the poor filtering of the DC/DC converter on the add-on board for PoDL power. When looking at the Add-On PoDL Schematic, there is only a 1 μ F capacitor as an input filter to the input pin of the converter. It may be possible to reduce the switching noise on the mixing segment by using a better filter.

Another important thing to note is that the type of noise is important. Depending on the network and line imbalances, any noise in an imbalanced system causes mode conversion, so that the common mode noise of the power supply may have an impact on the signal quality if the connections between boards and layouts of each board are not designed and managed correctly. In this case, radiated emissions may become an issue in a noise or EMI sensitive environment.

Additionally, these measurements were done in laboratory conditions at only 3.5W power usage at maximum. There was no dynamic load. The behavior of the mixing segment may look completely different at greater power usage or when dynamic loads are present on the network.

Finally, it is important to note a major advantage of using PoDL is to power remote nodes. While performing the measurements using PoDL, applying power to the individual nodes was no issue. It was possible by simply connecting the node to the mixing segment. However, when performing the measurements without PoDL, applying power to the individual nodes became an issue. Some of the USB cables were a different length than others, a USB hub had to be found and other complications arose due to the amount of cables required. It became very clear that the overhead for routing power to the required nodes may become an issue at a larger scale.

One easy takeaway from this is that while a mixing segment using PoDL becomes more susceptible to noise, this drawback may be heavily outweighed by the simplicity and ease of use when installing and powering the mixing segment. When no large power dynamics, low noise environments or low total power requirements are present for a mixing segment, it will generally be advantageous to use PoDL in the system and save effort on cabling and overhead.

7. Conclusions

After conducting these tests, the final step is to analyze the results. The comparison between the simulations and measurements showed that the results are nearly identical. In general, the simulations predicted that the system would behave worse than what was seen during measurements. Therefore, if the simulations show that the system performs within the acceptable range, then the real system will function as well. The system simulations and measurements showed that a fully functional system using Power over Data Line is possible.

As a reference, the system performance for all topologies was compared to the performance without using PoDL. This was also done for the eight node system. The results of these tests were compared to the corresponding system running PoDL. The result of this comparison showed that the system behavior of the systems using PoDL are very similar to the systems not using PoDL. The systems using PoDL are more susceptible to alien noise, but only by a small margin. This shows that much of the inherent noise of a system stems from the topology and interfaces to the network, both of which affect the system regardless.

While all topologies and component selections that were tested stayed within the required noise margin, clear advantages and disadvantages of each power coupling filter type were discovered. For the filter based on uncoupled single inductors, the resulting system has the lowest noise margin from the filters that were tested. This value degrades even further the lower the value of the inductance, the higher the distance, and the larger the node count become. Filters based on coupled inductors offer a good balance between costs, layout size and noise margin. Coupled inductors have a larger noise margin than separate inductors, but the best performance can be achieved using a filter based on DMCs. Systems using a filter based on DMCs will have the largest noise margin and will therefore be more robust against external noise effects. However, when a large current is required the size and cost of the DMC will greatly increase.

The simulations and tests confirm that DMCs are the most suitable components tested if a large noise margin is required. The tests show that the noise behavior of the system is very close to the behavior without the use of PoDL. However, the simulations and tests showed that some care must be taken in regards to the selection of components. For the power supply filter on each device, at least 80 μH of inductance in total are recommended. For single inductors, this means at least 40 μH on each of the positive and negative wires of the network. The PSE should also support the total power requirements of the system, potentially requiring inductors that have a higher rated current than the powered devices on the network.

Since the possible combinations of power supply voltages, regulators, coupling topology, nominal inductor value and variance, system topology and component vendors is nearly infinite, it is impossible to clearly say which combinations and setups will work. It is vital that simulations are used to assure the functionality of the desired system. This application note shows that with the right model, an entire system can be simulated such that the resulting measurements of a real system are the same as the simulations.

Furthermore, the systems shown here can be viewed as a performance floor. The examples shown here are still functional, so if a new system adheres to the electrical characteristics of its components and the simulations show similar performance to the results presented here, then this new system will also be functional.

An important conclusion to draw from this example is that system simulation is invaluable. It is imperative that the total current and voltage of each component is not exceeded and that the total system noise margin is not wholly consumed by the inherent system noise. The required margin for external noise depends largely on the expected external noise and environmental factors, if the system is expected to be subject to much alien noise then the simulations and topology of the system need to show a larger noise margin for the system to still work in that environment. Early simulation helps detect flaws in the design or topology of a desired system and allows to implement changes early in the design phase.

8. Revision History

Table 8-1. Revision History

Revision Level & Date	Section/Figure/Entry	Correction
DS60001848B (Oct-2025)	Chapter 5: PoDL Evaluation Measurements	Corrected signal shape of generated noise
	Chapter 6: Example Applications	Initial
DS60001848A (June-2024)	All	Initial

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